

Matrix Exponential: Given a square matrix A the matrix exponential, e^{tA} , can be calculated by diagonalizing A into $A = PDP^{-1}$ such that if A has eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ with corresponding eigenvectors $u_1, u_2, \dots, u_n$ then

$$e^{tA} = Pe^{tD}P^{-1} = \begin{pmatrix} u_1 & u_2 & \cdots & u_n \end{pmatrix} \begin{pmatrix} e^{t\lambda_1} & 0 & \cdots & 0 \\ 0 & e^{t\lambda_2} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & 0 & e^{t\lambda_n} \end{pmatrix} \begin{pmatrix} u_1 & u_2 & \cdots & u_n \end{pmatrix}^{-1}$$

If A has a repeated eigenvalue λ , with corresponding eigenvector v_1 , then the generalized eigenvector can be found by solving $(A - \lambda I)^k v_2 = v_1$ for some k . Standardly we have $(A - \lambda I)v_2 = v_1$ where v_2 is the generalized eigenvector. As well, the Jordan form will change to

$$J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \Rightarrow e^{tJ} = \begin{pmatrix} e^{\lambda t} & te^{\lambda t} \\ 0 & e^{\lambda t} \end{pmatrix}$$

We also have that for

$$\mathbf{x}'(t) = A\mathbf{x}(t), \quad \mathbf{x}(0) = \mathbf{x}_0$$

the general solution is

$$\mathbf{x}(t) = e^{tA}\mathbf{x}_0$$

with e^{tA} the matrix exponential.

Duhamels Formula: For the non-autonomous equation

$$\mathbf{x}'(t) = A\mathbf{x}(t) + \mathbf{g}(t)$$

Duhamels Formula provides the general solution:

$$\mathbf{x}(t) = e^{(t-t_0)A}\mathbf{x}_0 + \int_{t_0}^t e^{(t-s)A}\mathbf{g}(s) \, ds$$

for the initial value $\mathbf{x}(t_0) = \mathbf{x}_0$.

Picard Iteration: For an IVP with

$$x'(t) = v(x(t), t), \quad x(t_0) = x_0$$

define

$$X_0 = x_0, \quad X_1 = x_0 + \int_{t_0}^t v(x_0(s), s) \, ds, \dots, \quad X_n = x_0 + \int_{t_0}^t v(x_{n-1}(s), s) \, ds.$$

The solution to the IVP is then $\lim_{n \rightarrow \infty} X_n$

Supremum Distance: For f, g continuous functions on $[a, b]$ define

$$d(f, g) = \max_{t \in [a, b]} |f(t) - g(t)|.$$

Tip: Square the distance to remove the absolute value bars

Second Order ODEs: For an equation in the form

$$ax''(t) + bx'(t) + cx(t) = f(t)$$

start by solving the homogenous equation where $f(t) = 0$ and $a, b, c \in \mathbb{R}$, with the characteristic equation $ar^2 + br + c = 0$.

• If $r_1 \neq r_2$ then $x_h = C_1 e^{r_1 t} + C_2 e^{r_2 t}$

• If $r_1 = r_2$ then $x_h = (C_1 + C_2 t) e^{r_1 t}$

Then solve for the particular solution by guessing:

• $g(x) = e^{\alpha t} \Rightarrow x_p = C e^{\alpha t}$

• $g(x)$ is an n -degree polynomial $\Rightarrow x_p = A_n t^n + \cdots + A_1 t + A_0$ (can omit constant term based on context)

• $g(x) = A \cos bt$ or $A \sin bt \Rightarrow x_p = C \cos bt + D \sin bt$.

If the x_p guess is already accounted for in the homogenous solution, multiply by t until it is unique. The final solution is then $x = x_p + x_h$.

Cauchy-Euler Equations: For an equation in the form

$$t^2 x'' + atx' + bx = g(t)$$

guess $x_1 = t^\alpha$ for the homogenous equation so that $x_1' = \alpha t^{\alpha-1}$ and $x_1'' = \alpha(\alpha-1)t^{\alpha-2}$. It follows that

$$\alpha(\alpha-1) + a\alpha + b = 0$$

• If $\alpha_1 \neq \alpha_2$ then $x_h = C_1 t^{\alpha_1} + C_2 t^{\alpha_2}$

• If $\alpha_1 = \alpha_2$ then $x_h = C_1 t^{\alpha_1} + C_2 t^{\alpha_1} \ln t$

Then find the particular solution by guessing or variation of parameters. The final solution is then $x = x_1 + x_p$.

Variation of Parameters: If $x_1(t)$ and $x_2(t)$ are solutions to the homogenous equation then a particular solution is

$$x_p = \int_{t_0}^t \frac{\det(N(t, s))}{\det(M(s))} f(s) ds$$

for

$$N(t, s) = \begin{pmatrix} x_1(s) & x_2(s) \\ x_1(t) & x_2(t) \end{pmatrix} \quad \text{and} \quad M(s) = \begin{pmatrix} x_1(s) & x_2(s) \\ x_1'(s) & x_2'(s) \end{pmatrix}$$

Solution Space Dim: A n th order differential equation has a solution space with order n . Thus, any solution set with more than n solutions, must contain at least one linearly dependent solution.

Inner Product Spaces: A function f defined on $[a, b]$ is in $L^2[a, b]$, the inner product space of all square integrable functions, if, and only if,

$$\int_a^b |f(x)|^2 dx < \infty$$

Approximation in Inner Product Spaces: To find coefficients $c_0, c_1, c_2, \dots, c_k$ to approximation some function f in $L^2[a, b]$ given a set of basis functions $\{\phi_1, \phi_2, \dots, \phi_k\}$, we must minimize

$$\|f - (c_0 + c_1\phi_1 + c_2\phi_2 + \dots + c_k\phi_k)\|^2 = \int_a^b (f - c_0 - c_1\phi_1 - c_2\phi_2 - \dots - c_k\phi_k)^2 dx.$$

We can find the smallest such c_k by computing the projection:

$$c_k = \frac{\langle f, \phi_k \rangle}{\langle \phi_k, \phi_k \rangle}$$

where the inner product $\langle f, g \rangle = \int_a^b f(x)g(x) dx$ provided the basis functions are all orthogonal (i.e. $\langle f, g \rangle = \int_a^b f(x)g(x) dx = 0$ for all basis functions).

Good to know formulas:

$$e^t = \sum_{k=0}^{\infty} \frac{t^k}{k!} = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots$$

$$\sin t = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} t^{2k+1} = t - \frac{t^3}{3!} + \frac{t^5}{5!} + \dots$$

$$\sin 2t = 2 \sin t \cos t$$

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$te^t = \sum_{k=0}^{\infty} \frac{t^{k+1}}{k!} = t + t^2 + \frac{t^3}{2!} + \frac{t^3}{3!} + \dots$$

$$\cos t = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} t^{2k} = 1 - \frac{t^2}{2!} + \frac{t^4}{4!} + \dots$$

$$\cos 2t = \cos^2 t - \sin^2 t$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \pm \sin \alpha \sin \beta$$

$$\frac{1}{1-t} = \sum_{t=0}^{\infty} t^n = 1 + t + t^2 + \dots$$

$$\cos^2(x) = \frac{1 + \cos(2x)}{2}$$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\sin^2(x) = \frac{1 - \cos(2x)}{2}$$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \lambda^2 - (a+d)\lambda + (ad-bc) = 0$$